Pre-Analysis Plan: Effects of Having Daughters on Gender Attitudes, Political Preferences, and Household Equality

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Overview and Main Sample Restriction

This pre-analysis plan outlines the methodology to test four hypotheses about the effects of having daughters on attitudes, political preferences, and gender equality behaviors. For all four hypotheses, we restrict the analysis to respondents who are parents. Specifically, we include all men and women in the available SOEP data across all observable years who have at least one child. Individuals who have never had children at the time of analysis are excluded, so that all estimates compare parents with at least one daughter to parents without daughters.

We identify parents and the gender composition of their children using the `biol` dataset[1], which records every child ever reported by a survey respondent, including the child's gender and year of birth. This allows us to classify respondents into those who have at least one daughter and those who are parents but do not have a daughter at the time of the specific observation. This classification forms the basis of our main treatment variable: a binary indicator for having a daughter. Note that this means a respondent who first has a son and later a daughter will initially count as untreated and become treated at the time of the birth of the daughter. Sub-samples for each hypothesis will be further defined according to the availability of the relevant outcome variables and the specifics of the hypothesis tested.

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[1] `biol` is a long-format dataset containing cumulated individual-level raw data from the biographical questionnaire and from wave-specific biographical modules of the individual questionnaire. It provides information on every child reported by a respondent for up to ten children, including sex and date of birth harmonized across all existing SOEP waves.

We will for each regression listed below run three specifications: 1) in which we will not include any controls; 2) in which we will include controls for key demographic characteristics of the respondent, including age, gender, education, migration background, East/West Germany indicator, employment variables (where this is not the outcome), and marital status. We will also control for the total number of children of the respondent at the time of the interview, as well as the number of children below age 16, to account for potential confounding by family structure. In the final specification 3), we will additionally control for the age and migration status of the partner and, again, when it is not the outcome, the employment variables of the partner as well as the educational background of the parents.

Data Description

The analysis will be conducted using data from the German Socio-Economic Panel (SOEP), version 2020 (Goebel et al., 2019; Siegers, Steinhauer, and Schuett, 2022). The SOEP is a longitudinal panel study of private households in Germany, collecting annual data since 1984. For this project, we use data from 2000 to 2020, covering all individuals who completed the individual questionnaire in at least one of these years. The panel is unbalanced.

The final panel dataset is constructed by merging files using the personal identifier (`pid`), the survey year identifier (`syear`), and, when needed to identify partners, the household ID (`hid`) and the partner ID (`pgpartnr` from dataset `pgen`) as unique keys.

Our dataset includes only recoded and newly computed variables (e.g., age of children at the time of the interview, gross hourly wage), created by combining information from the provided datasets.

As mentioned above, we will always run three specifications, for any subsample we define in the respective hypotheses section:

1. no controls
2. core controls (vary by hypotheses as defined in Table 1)
3. core controls + additional controls (vary by hypotheses as defined in Table 1)

Table 1 below provides an overview of all variables that are not hypothesis specific, their dataset source and the source variables, as well as a description of how we intend to compute/generate them. In each hypothesis section, we will further describe in more detail the respective outcome variables of interest, as well as the subsample decision rules for including/excluding observations given data limitations or the nature of the hypothesis tested.

Table 1: Overview of variables, sources, and computation

Variable	Description	Source dataset	Source variable(s)	Computation / Recoding
Core controls for specifications 2+3, H1-4				
age	Age of respondent	ppathl	syear, gebjahr	Subtract year of birth from survey year to compute age in years
male	Gender of respondent	ppathl	sex	Create dummy: 1 if male, 0 if female
migback	Direct migration background	ppathl	migback	Code as 1 if direct migration background, 0 otherwise
east	East Germany indicator	ppathl	sampreg	Code as 1 if living in East Germany, 0 if West Germany
numchild	Total number of children at the time of interview	biobirth	kidgeb01-kidgeb19, syear	Count the number of children whose birth year was prior to the year of the interview
childu16	Total number of children under age 16	biobirth	kidgeb01-kidgeb19, syear	Compute age for each child as survey year minus year of birth; count how many children of respondent are younger than 16 in survey year
Core controls for specification 2+3, H1,H2+H4				
lfs	Employment status of respondent	pgen	pgemplst	Categorize into not employed, not employed
brstdlov	Gross hourly wage	pgen	pglabgro, pgvebzeit	Calculate gross wage per hour; clean by trimming extreme values
workhours	Weekly work hours	pgen	pgvebzeit	Use reported contractual work hours per week

Table 1 Continued from previous page

Variable	Description	Source dataset	Source variable(s)	Computation / Recoding
Additional controls for specification 3, all hypotheses				
father_abi	Father has high-school diploma	bioparen	fsedu	Code as 1 if father has at least highschool diploma and 0 otherwise
mother_abi	Mother has high-school diploma	bioparen	msedu	Code as 1 if mother has at least highschool diploma and 0 otherwise
Additional controls for specification 3 only, H1,H2+H4				
age_p	Age of respondent's partner	ppathl	syear, gebjahr	Subtract year of birth from survey year to compute age in years
migback_p	Direct migration background of partner	ppathl	migback	Code as 1 if direct migration background, 0 otherwise
lfs_p	Employment status of respondent's partner	pgen	pgemplst	Categorize into employed, not employed
brstdlov_p	Gross hourly wage of partner	pgen	pglabgro, pgvebzeit	Calculate gross wage per hour; clean by trimming extreme values
workhours_p	Weekly work hours of partner	pgen	pgvebzeit	Use reported contractual work hours per week
Main independent variable				
has_daughter	Respondent has daughter(s)	biobirth	kidsex01-kidsex19, kidgeb01-kidgeb19, syear	Code as 1 if any child is female and born before survey year, 0 otherwise

Daughter indicator and household composition: A respondent is classified as having a daughter if any child is recorded as female and born before the interview year (syear).

The indicator remains 1 once set.

Decision rule for being included in any sample:

We will keep all respondents who have any child at the time of the interview, based on our computed `numchild` variable (i.e., if this is larger than 0). We will only keep respondents whose age is between 18 and 65 at the time of the interview, as computed with our `age` variable.

Finally, for all variables, we will recode them to missing in the following way:

```
mvdecode _all, mv(-1=. \ -2=.a \ -3=.c \ -4=.d \ -5=.b \
-6=.e \ -7=.f \ -8=.g \ -9=.h \ -10=.i)
```

Legend:

- . = No answer (-1)
- .a = Does not apply (-2)
- .b = Not included in questionnaire version (-5)
- .c = Not valid (-3)
- .d = Forbidden multiple response (-4)
- .e = Questionnaire version with modified filter (-6)
- .f = Only available in less restricted edition (-7)
- .g = Question not included this year (-8)

1 Gender Equality Attitudes (X)

Hypothesis 1 *Respondents with daughters show stronger support for gender equality (X) than those without daughters.*

The SOEP is a rich longitudinal dataset with extensive coverage of socio-economic and demographic information. However, it has limited coverage of gender attitude measures. Questions relevant to gender role attitudes were asked only in two waves: 2012 and 2018. Even then, not all items were repeated across both years, making it difficult to construct a consistent panel measure of gender attitudes. It represents a notable gap in an otherwise

comprehensive panel and points to an opportunity to strengthen future SOEP waves by including a broader and more consistent battery of gender equality attitude questions.

Given variable availability and design trade-offs, we propose three main empirical strategies for this hypothesis. The first offers broader question coverage, the second offers broader sample coverage, while the third has stronger causal identification. We plan to prioritize Analyses 1 and 2 for feasibility, but provide all three strategies below.

Variables

Our analysis relies on attitude items included in the individual questionnaire (`pl` dataset), specifically from the 2012 and 2018 waves. Each question is measured on a 5-point Likert scale. Items are reverse-coded where necessary so that higher values consistently reflect more support for gender equality (see Table 2). We will then standardize each item (z-score), average across available items and generate our main dependent variable for Hypotheses 1 and 4, gender_attitudes_index.

Table 2: Gender Attitude Items and Coding

Variable	**Description**	**Source dataset**	**Source variable(s)**	**Computation / Recoding**
momwork	Child under 3 suffers if mother works	`pl`	`plh0302_v1` (2012) `plh0302_v2` (2018)	Reverse coded
equalshare	Best if man and woman work same amount	`pl`	`plh0308_v1` (2012) `plh0308_v2` (2018)	No
careerpriority	Women should care more about family than career	`pl`	`plh0301` (2012)	Reverse coded
housework	Men involved in housework	`pl`	`plh0303` (2012)	No
fathercareer	Children suffer when father focused on career	`pl`	`plh0304` (2012)	No

Table 2 Continued from previous page

Variable	Description	Source dataset	Source variable(s)	Computation / Recoding
menmoney	Men care more about financial security	`pl`	`plh0307` (2012)	Reverse coded
warmmother	Working mothers equally emotionally warm	`pl`	`plh0309`	No

Analysis 1 & 2 – Cross-sectional (OLS)

Goal: Compare gender attitudes of the respondents with only sons versus those with at least one daughter, using a cross-sectional design.

Samples: Two possible samples will be employed:

- **2012 version:** Individuals who had at least one child by 2012 and responded to the 2012 gender attitude module, 7 questions.
- **2012 & 2018 combined:** Individuals who had at least one child by the year of interest and responded to the 2 questions available both in 2012 and 2018.

Treatment: Indicator for having at least one daughter (versus only sons).

Inclusion/Exclusion Criteria

- **Include:** Respondents with at least one biological child by the year of interest (2012 or 2018), and who completed the relevant gender attitude items.
- **Exclude:** Respondents without children and who have not completed the items.

Exact Specification (OLS)

$$\texttt{gender_attitudes_index}_i = \beta_0 + \beta_1 \texttt{has_daughter}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \epsilon_i$$

- `has_daughter`: 1 if respondent has at least one daughter
- $\mathbf{X}_i$: Control vector (see Table 1)

- Standard errors: Robust

Interpretation

- β_1 captures average differences in gender equality attitudes between parents with and without daughters.
- A positive and statistically significant (at 5% level) β_1 supports the hypothesis.

Analysis 3 – Panel (Difference-in-Differences, Causal)

Goal: Estimate the causal effect of having a daughter on gender attitudes using a DiD design.

Treatment group: Respondents who had at least one daughter between 2012 and 2018.

Control group: Respondents who had at least one son but no daughters during the same period.

Note: This approach allows causal identification *if* sample sizes are sufficient.

Inclusion/Exclusion Criteria

- **Include:** Respondents who had at least one child born between 2012 and 2018, who had only sons before 2012 (or, depending on the data availability, no children), and who answered the 2012 and 2018 gender attitude items.
- **Exclude:** Respondents with no children born between 2012 and 2018 or who have not answered gender attitude items.

Exact Specification (Difference-in-Differences)

$$\begin{aligned}\texttt{gender_attitudes_index}_{it} = \beta_0 + \beta_1 \texttt{has_daughter}_i \times \texttt{Post2018}_t \\ + \beta_2 \texttt{has_daughter}_i + \beta_3 \texttt{Post2018}_t + \mathbf{X}_i' \boldsymbol{\gamma}_1 + \mathbf{Y}_{it}' \boldsymbol{\gamma}_1 \epsilon_{it}\end{aligned}$$

- `has_daughter`: 1 if respondent had at least one daughter between 2012 and 2018
- `Post2018`: 1 if wave = 2018
- $\mathbf{X}_i$: Time-invariant controls

- $\mathbf{Y}_{it}$: Time-varying controls
- Standard errors: Clustered at individual level

Interpretation

- β_1 is the difference-in-differences estimator.
- A positive and statistically significant (at 5% level) β_1 indicates that having a daughter causes a greater increase in gender equality attitudes than having a son.

2 Political Preferences (Y)

Hypothesis 2 *Respondents with daughters exhibit more left-leaning political preferences (Y).*

The German Socio-Economic Panel (SOEP) includes several variables relevant to political preferences. We focus on two indicators: ideological self-placement on a left–right scale and stated party affiliation. These are used as two complementary outcome variables in our analysis (see Table 3).

Variables

We treat political tendency as our main outcome and use party affiliation (converted to an ideology score) to validate and extend findings.

Table 3: Political Preferences Items and Coding

Variable	**Question**	**Source dataset**	**Source variable(s)**	**Computation / Recoding**
leftright	Left–right political tendency	`pl`	`plh0004`	Respondents rate their political views on a scale from 0 (far left) to 10 (far right). Available in 2005, 2009, 2014, and 2019. We reverse-code this variable so that higher values reflect more left-leaning views.

Table 3 Continued from previous page

Variable	Description	Source dataset	Source variable(s)	Computation / Recoding
party	Party affiliation	`pl`	`plh0012_h`	Available in all years between 1984 and 2020. Respondents indicate the political party they identify with. We will only keep single mentions of parties and recode the mixed-party answers as missings, except CDU/CSU as we will combine both in one party. We match party codes with ideological scores from the WZB Manifesto Project, which locates each party on a left–right continuum for repeated years over our observed time period. We will impute this measure for all years using a backward-fill method around the available years from the MANIFESTO data to cover the whole period from 2000-2020. We then assign the respective left-right value for each party-year combination to the parties listed in the SOEP in each year. This enables a continuous left–right scale suitable for OLS analysis.

Treatment

Indicator for having at least one daughter (versus only sons).

Inclusion/Exclusion Criteria

- **Include:** Respondents with at least one child and valid responses to political tendency or party affiliation.
- **Exclude:** Respondents without children or with missing political preference data.

Exact Specification

We estimate the following model:

$$\texttt{political_pref}_{it} = \beta_0 + \beta_1 \texttt{has_daughter}_{it} + \mathbf{X}_i'\boldsymbol{\gamma} + \mathbf{Y}_{it}'\boldsymbol{\delta} + \tau_t + \epsilon_{it}$$

- `political_pref`: Political preference of individual i in year t, either ideological self-placement or party-based ideology score
- `has_daughter`: 1 if individual has at least one daughter
- $\mathbf{X}_i$: Time-invariant controls
- $\mathbf{Y}_{it}$: Time-varying controls
- Standard errors: Clustered at the individual level
- We include time fixed effects using `i.syear` to control for common shocks and period-specific influences across survey waves.

Interpretation

- β_1 captures the average association between having a daughter and reporting more left-leaning political preferences.
- A positive and statistically significant (at 5% level) β_1 supports the hypothesis.

3 Household Gender Equality

Hypothesis 3 *Having daughters leads to more gender-equal behaviors or outcomes in the household.*

We measure gender-equal behaviors in the household by creating within-household indicators of the division of paid and unpaid work. Our focus is on the actual division of labor

within different-sex couples, as identified in the SOEP. We therefore restrict the sample to participants for whom we can link both partners via `pgpartnr` and confirm they share the same household ID (`hid`). We construct a couple dataset by saving female records (suffix _f) and male records (suffix _m), and then merging these based on partner linkage (pgpartnr) and household id (hid). We only keep those for which we have information from both partners living in one household on all variables. All time-invariant and time-varying control variables are included for both partners, identified by the suffixes _f (female) and _m (male). For some controls — specifically `east`, `numchild`, and `childu16`, we only use the female partner's value (_f).

From this couple sample, we compute household-level measures capturing the relative contributions of each partner along several dimensions: hourly wage, contractual work hours, housework hours, and childcare hours. The dependent variables are constructed as gaps representing the relative share of the respective female to the male partner.

Variables

For the paid work measure, we use the variables `brstdlov` and `workhours` described in Table 1 for the female respondents. As with all generated variables in the couple dataset, they are renamed with the suffix _f, resulting in `brstdlov_f` and `workhours_f`. We then merge the corresponding information for the partner from the stored male dataset, which provides `brstdlov_m` and `workhours_m`.

We further include measures of the within-household gender care gap in childcare and housework. Table 4 below describes the variables generated in the full sample for this purpose:

Table 4: Household time use variables (weekday)

Variable	**Description**	**Source dataset**	**Source variable(s)**	**Computation / Recoding**
childcare_wd	Childcare hours on a weekday	`pl`	`pli0044_h`	Computed from reported weekday childcare hours.
housework_wd	Housework hours on a weekday	`pl`	`pli0043_h`, `pli0040`, `pli0049_h`	Sum of hours for housework, errands, and repairs on a weekday.

In the merged different-sex partner sample, we will, as all generated variables, have this

measure for the female partner with the suffix _f as well as the male partner, with suffix _m.

We compute the following relative measures:

- **Gender pay gap (GPG)**:

$$\texttt{gpg} = \frac{\texttt{wage_m} - \texttt{wage_f}}{\texttt{wage_m}} \times 100$$

  GPG in percent, relative to men's wage.

- **Gender work hours gap (GWG)**:

$$\texttt{gwg} = \frac{\texttt{workhours_m} - \texttt{workhours_f}}{\texttt{workhours_m}} \times 100$$

  GWG in percent, relative to men's work hours.

- **Gender childcare gap (GCCG childcare)**:

$$\texttt{gccg_childcare} = \frac{\texttt{childcare_wd_f} - \texttt{childcare_wd_m}}{\texttt{childcare_wd_f}} \times 100$$

  GCCG childcare in percent, relative to hours women spend on childcare.

- **Gender housework gap (GHG housework)**:

$$\texttt{ghg_housework} = \frac{\texttt{housework_wd_f} - \texttt{housework_wd_m}}{\texttt{housework_wd_f}} \times 100$$

  GHG housework in percent, relative to hours women spend on housework.

Treatment

Indicator for having at least one daughter (versus only sons).

Inclusion/Exclusion Criteria

- **Include:** Heterosexual couples with linked partner records (shared `hid` and `pgpartnr`), with valid information for both partners on the relevant variables.
- **Childcare regression:** Only include couples with at least one child under the age of 16 living in the household.
- **Exclude:** Couples without valid data for either partner on key variables.

Exact specification (OLS)

We estimate the following model:

$$Y_{ct} = \beta_0 + \beta_1 \texttt{has_daughter}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \mathbf{Z}_{it}'\boldsymbol{\delta} + \tau_t + \epsilon_{ct}$$

- $\mathbf{Y}_{ct}$: Household gender gap outcome of couple c in year t (GPG, GWG, GCG childcare, or GCG housework)
- `has_daughter`: 1 if individual has at least one daughter
- $\mathbf{X}_i$: Time-invariant controls
- $\mathbf{Z}_{it}$: Time-varying controls
- Standard errors: Clustered at the individual level
- We include time fixed effects using `i.syear` to control for common shocks and period-specific influences across survey waves.

Time-invariant and time-varying controls are included for both partners (suffix _f and _m), except for `east`, `numchild`, and `childu16`, which are only included for the female partner.

Interpretation

- β_1 captures the average association between having a daughter and the size of the respective within-household gender gap.
- A statistically significant (at 5% level) β_1 supports the hypothesis that having a daughter is associated with a smaller gender gap within the household.

4 Gender Equality Attitudes Between Sexes (X)

Hypothesis 4 *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*

This hypothesis builds directly on Hypothesis 1, using the same data sources, attitude measures, and empirical strategies, but focuses on whether the effect of having daughters on gender equality attitudes differs by the respondent's sex. Specifically, we examine whether the attitudinal shift is larger among men than among women.

As in Hypothesis 1, gender attitude items are only available in the 2012 and 2018 waves, and even then, only partially overlapping across years. Given these constraints, we again propose two empirical strategies. The first prioritizes feasibility and broader sample coverage, while the second offers stronger causal identification. The core structure mirrors Hypothesis 1, with the key addition of a male dummy and its interaction with the treatment. In the panel framework (Analysis 2), we extend the approach to a triple-difference (DDD) design, conditional on sufficient sample size.

Variables

See Hypothesis 1.

Analysis 1 & 2 – Cross-sectional (OLS with Interaction)

Goal: Compare whether the difference in gender attitudes between respondents with and without daughters is larger among men than women.

Samples: Two possible samples could be employed:

- **2012 version:** Individuals who had at least one child by 2012 and responded to the 2012 gender attitude module, 7 questions.
- **2012 & 2018 combined:** Individuals who had at least one child by the year of interest and responded to the 2 questions available both in 2012 and 2018.

Treatment: Interaction term between having at least one daughter and being male.

Inclusion/Exclusion Criteria

- **Include:** Respondents with at least one biological child by the year of interest (2012 or 2018), and who completed the relevant gender attitude items.
- **Exclude:** Respondents without children and who have not completed the items.

Exact Specification (OLS)

$$\begin{aligned}\texttt{gender_attitudes_index}_i = \beta_0 + \beta_1(\texttt{has_daughter}_i \times \texttt{male}_i) \\ +\beta_2\texttt{has_daughter}_i + \beta_3\texttt{male}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \epsilon_i\end{aligned}$$

- `has_daughter`: 1 if respondent has at least one daughter
- $\mathbf{X}_i$: Control vector (see Table 1)
- Standard errors: Robust

Interpretation

- β_1 captures whether the daughter effect is significantly different for men versus women.
- A positive and statistically significant (at 5% level) β_1 supports the hypothesis.

Analysis 3 – Panel (Triple Difference-in-Differences, Causal)

Goal: Estimate whether having a daughter causes a greater increase in gender equality attitudes for men than for women using a DDD design.

Treatment group: Respondents who had at least one daughter between 2012 and 2018.

Control group: Respondents who had at least one son but no daughters during the same period.

Note: This design allows causal identification of heterogeneous treatment effects by gender, but is tentative due to possible sample size limitations.

Inclusion/Exclusion Criteria

- **Include:** Respondents who had at least one child born between 2012 and 2018, who had only sons before 2012 (or, depending on the data availability, no children), and who answered the 2012 and 2018 gender attitude items.
- **Exclude:** Respondents with no children born between 2012 and 2018 or who have not answered gender attitude items.

Exact Specification (Difference-in-Differences-in-Differences)

$$\begin{aligned}\texttt{gender_attitudes_index}_{it} = \beta_0 + \beta_1 \texttt{has_daughter}_i \times \texttt{Post2018}_t \times \texttt{male}_i \\ + \beta_2 \texttt{has_daughter}_i \times \texttt{Post2018}_t + \beta_3 \texttt{has_daughter}_i \times \texttt{male}_i \\ + \beta_4 \texttt{Post2018}_t \times \texttt{male}_i + \beta_5 \texttt{has_daughter}_i + \beta_6 \texttt{Post2018}_t + \beta_7 \texttt{male}_i \\ + \mathbf{X}_i' \boldsymbol{\gamma}_1 + \mathbf{Y}_{it}' \boldsymbol{\gamma}_1 + \epsilon_{it}\end{aligned}$$

- `has_daughter`: 1 if respondent had at least one daughter between 2012 and 2018
- `Post2018`: 1 if wave = 2018
- `male`: 1 if respondent is male
- $\mathbf{X}_i$: Time-invariant controls
- $\mathbf{Y}_{it}$: Time-varying controls
- Standard errors: Clustered at individual level

Interpretation

- β_1 is the triple-difference estimator.
- A positive and statistically significant (at 5% level) β_1 would support the hypothesis that the attitudinal shift is stronger for men than for women after having daughters.